

Application Flow Document
Eila Eco Pencils E-Commerce Platform

1. Overview

This document describes the end-to-end application flow for customers, administrators, payment processing, inventory updates, and supporting modules.

Flow Notes

Each flow should include decision points, validation, loading states, success and failure paths, security checks, and user feedback. Sequence diagrams and BPMN/UML activity diagrams can be derived from these flows during implementation.

2. Customer Flow

Landing Page → Browse Products → Product Details → Add to Cart → Cart → Checkout → Shipping Address → Razorpay Payment → Payment Verification → Order Success → WhatsApp Confirmation

Flow Notes

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3. Admin Flow

Admin Login → Dashboard → Products → Add/Edit/Delete Products → Inventory Management → Orders → Update Status → Analytics → Bulk Inquiries → Logout

Flow Notes

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4. Payment Flow

Checkout → Backend creates Razorpay Order → Customer Pays → Razorpay Callback → Backend Signature Verification → Store Payment → Create Order → Reduce Stock → Success Page

Flow Notes

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5. Inventory Flow

Admin Adds Product → Product Published → Customer Purchases → Stock Reduced → Low Stock Alert → Admin Restocks → Inventory Log Updated

Flow Notes

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6. Corporate Inquiry Flow

Corporate Page → Fill Inquiry Form → Submit → Store Inquiry → Admin Reviews → Contact Customer → Convert to Order

Flow Notes

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7. Error Handling Flow

Validation Error → Display Friendly Message → Retry → Server Error → Log Error → Notify Admin → Recovery

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8. Future Flow

Optional Account Creation → Order History → Wishlist → Reviews → Loyalty → AI Recommendations

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High-Level Flow Summary

Module	Input	Output
Homepage	Visitor	Product Discovery
Product	Selection	Cart Item
Cart	Products	Checkout
Checkout	Shipping Details	Payment
Payment	Razorpay	Verified Order
Orders	Verified Payment	Order Tracking
Admin	Management Actions	Updated Store